Elton Chang

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Hiring Team
Competition
The Brattle Group
London, United Kingdom

Dear Hiring Team,

I am applying for the 2027 Research Analyst position in Competition in your London office. I completed an M.S. in Data Analytics for Science at Carnegie Mellon University in May 2026, with a 3.97 GPA, and a B.S. in Computer Science at the University of California, Santa Cruz. I want to join a project team where quantitative work is tied to a real question: imperfect data, a model that has to be checked, and a result a colleague can explain.

In Quantifying Meritocracy, I extended the University of Catania's Talent vs. Luck model with a solidarity parameter. I used agent-based simulation to compare redistribution strategies on inequality, upward mobility, and wealth concentration, and I ran sensitivity analyses across policy timing and model parameters. I documented the results as comparisons produced by the model. The habit I would bring to project work is the same one: name the mechanism, choose a metric that matches the question, and keep the comparison inside what the model can support.

I have also built and checked models on operational records. At Ashe System, where I am founder and technical lead, I ranked aircraft-on-ground risk from operators' maintenance, inventory, and supplier exports. Using XGBoost, validated Precision@20 improved from 35% to 62%. Tuning parameters and decision thresholds with Optuna increased median warning lead time from 6.5 to 10.2 days. Each recommendation keeps the evidence behind it, and the product is decision support for human review. In a Carnegie Mellon collaboration with PNC, I compared histogram gradient boosting, LightGBM, XGBoost, and an ensemble for next-month product adoption. The pipeline reached 0.811 ROC AUC, and a supported recommendation appeared in the top three 95% of the time. We reported both numbers because a ranking is useful when the short list a person reviews contains the relevant action.

The PNC analysis was a team project, and as a software engineer intern at Wintec Industries I wrote architecture and workflow documentation that improved onboarding and platform adoption by 35%. I revise an analysis so a teammate can rerun it, and I write down where the data stop. I do this work in Python, with SQL when the records are relational.

I would like to learn Competition research with Brattle's London team: how a market question becomes a measurement, how a model is checked before it is relied on, and how the finding is written so a colleague can use it.

Thank you for considering my application. I would welcome a conversation.

Sincerely,

Elton Chang